

How It Works

The program loads the data for the maze's walls from a text file and into a dictionary stored in the `maze` variable. This dictionary has (x, y) tuples for keys and the string in the `WALL`, `EMPTY`, `START`, or `EXIT` constants for values. Project 45, "Maze Runner 3D," uses a similar dictionary representation of the maze. The difference between the projects is in the code that renders that maze on the screen. Since Maze Runner 2D is simpler, I recommend becoming familiar with this program first before moving on to Maze Runner 3D.

```
1. """Maze Runner 2D, by Al Sweigart al@inventwithpython.com
2. Move around a maze and try to escape. Maze files are generated by
3. mazemakerrec.py.
4. View this code at https://nostarch.com/big-book-small-python-projects
5. Tags: large, game, maze"""
6.
7. import sys, os
8.
9. # Maze file constants:
10. WALL = '#'
11. EMPTY = ' '
12. START = 'S'
13. EXIT = 'E'
14.
15. PLAYER = '@' # (!) Try changing this to '+' or 'o'.
16. BLOCK = chr(9617) # Character 9617 is '⬮'
17.
18.
19. def displayMaze(maze):
20.     # Display the maze:
21.     for y in range(HEIGHT):
22.         for x in range(WIDTH):
23.             if (x, y) == (playerx, playery):
24.                 print(PLAYER, end='')
25.             elif (x, y) == (exitx, exity):
26.                 print('X', end='')
27.             elif maze[(x, y)] == WALL:
28.                 print(BLOCK, end='')
29.             else:
30.                 print(maze[(x, y)], end='')
31.         print() # Print a newline after printing the row.
32.
33.
34. print("""Maze Runner 2D, by Al Sweigart al@inventwithpython.com
35.
36. (Maze files are generated by mazemakerrec.py)""")
37.
38. # Get the maze file's filename from the user:
39. while True:
40.     print('Enter the filename of the maze (or LIST or QUIT):')
41.     filename = input('> ')
42.
43.     # List all the maze files in the current folder:
```